PROJECT REPORT | CLOUD SECURITY

AWS–Azure Hybrid Cloud Connectivity Using Site-to-Site VPN

Private networking between Amazon VPC and Azure VNet

Clouds	AWS Mumbai (ap-south-1) and Azure Central India
Technologies	Site-to-Site VPN, IPsec / IKEv2, static routing
Report date	27 October 2026
Prepared by	_____

Table of Contents

Table of Contents . . . 2
1. Executive Summary . . . 3
2. Objectives and Scope . . . 3
3. Network Design . . . 4
4. Solution Architecture . . . 5
5. Why Each Service Is Used . . . 6
6. Implementation Walkthrough . . . 8
7. Results Summary . . . 20
8. Project Notes Versus Screenshots . . . 21
9. Troubleshooting and Pending Work . . . 21
10. Findings and Recommendations . . . 23
11. Cost Management and Clean-up . . . 23
12. What Was Learned . . . 23
13. Resume-Ready Description . . . 24
14. Conclusion . . . 24
Appendix A – Resource Inventory . . . 25
Appendix B – Verification Commands . . . 25
Appendix C – Figure Index . . . 26
Appendix D – Glossary . . . 27

1. Executive Summary

This project connects a private network in **Amazon Web Services** with a private network in **Microsoft Azure** using a **Site-to-Site IPsec VPN**, so that resources in the two clouds can talk to each other over private IP addresses without exposing the test machines to the internet. The AWS side is an Amazon VPC (**AWS-VNET**, 10.1.0.0/16) in Mumbai (ap-south-1); the Azure side is a virtual network (**V-NET**, 10.2.0.0/16) in Central India. The two address ranges do not overlap, which is the first requirement for routing between them.

An AWS **Virtual Private Gateway** and an Azure **route-based VPN Gateway** (VpnGw1AZ) terminate the encrypted tunnel using **IKEv2**. Routes are exchanged **statically**: ROUTE-1 in AWS sends 10.2.0.0/16 to the Virtual Private Gateway, and the Azure Local Network Gateway **LNG-AWS** declares 10.1.0.0/16 as reachable through the tunnel. Security rules (an AWS security group and an Azure NSG) allow only ICMP between the two ranges for testing.

The tunnel came up (AWS **Tunnel 1: Up**, Azure connection **Connected**) and the Azure VM **VM-1** (10.2.1.4) successfully pinged the AWS EC2 instance (10.1.0.231) with **0% packet loss** and a round-trip time of about 6.3–6.5 ms. The reverse test (AWS to Azure), an application-level test, a second tunnel and Infrastructure as Code are still pending.

Item	Result
VPN type	Site-to-Site IPsec, IKEv2, static routing (no BGP), pre-shared key authentication
AWS tunnel status	Tunnel 1 Up, provisioning Available (status change 27 Oct 2026, 01:33:06 IST)
Azure connection status	Connected (Site-to-site IPsec)
Azure to AWS ping	Verified: 4 of 4 packets, 0% loss, average 6.530 ms (run 1) and 6.300 ms (run 2)
AWS to Azure ping	Not tested
Public IPs on test VMs	None (EC2 10.1.0.231 and VM-1 10.2.1.4 are private only)
Redundancy / IaC / HTTP test	Not implemented

2. Objectives and Scope

2.1 Objectives

- Connect an AWS VPC and an Azure VNet privately and securely with an encrypted Site-to-Site VPN.
- Plan non-overlapping address spaces and configure routing on both sides.
- Control traffic with security groups and network security groups, allowing only what the test needs.
- Prove connectivity with a private-IP ping between VMs that have no public IP addresses.
- Document the design, the reason each service is used, the limitations and the clean-up steps.

2.2 In scope

- AWS: VPC, subnet, route table, Virtual Private Gateway, Customer Gateway, Site-to-Site VPN connection, EC2 test instance, security group.
- Azure: resource group, VNet, VPN Gateway and public IP, Local Network Gateway, VPN connection, test VM, NIC and NSG.
- Connectivity verification from the Azure VM using the serial console.

2.3 Out of scope (pending)

- AWS-to-Azure ping, HTTP communication between the clouds, second tunnel / redundancy, BGP, and Infrastructure as Code (Terraform or Bicep).

3. Network Design

Component	AWS	Azure
Region	Asia Pacific (Mumbai), ap-south-1	Central India
Private network	VPC AWS-VNET (vpc-0ca4c93d6eab5c341)	VNet V-NET
Network CIDR	10.1.0.0/16	10.2.0.0/16
Subnet	Private-Subnet, ap-south-1a (10.1.0.0/24 per notes)	Private-Subnet (10.2.1.0/24 per notes) and GatewaySubnet
Test machine	EC2 i-093c6048486aa3590, t3.micro, 10.1.0.231	VM-1, Standard D2s v3, Ubuntu 24.04, 10.2.1.4
VPN gateway	Virtual Private Gateway V-gateway (vgw-095bf965a9b7c5f64), ASN 64512	VPN Gateway “VPN”, VpnGw1AZ, route-based
VPN endpoint IP	Tunnel 1 outside IP 16.4.101.178	Public IP “vpn” 4.187.151.232 (Standard SKU)
Remote-network object	Customer Gateway “Azure - gateway” (cgw-0fcefc590762d25eb)	Local Network Gateway LNG-AWS (10.1.0.0/16)
Route to the other cloud	ROUTE-1: 10.2.0.0/16 to Virtual Private Gateway	Via LNG-AWS address space 10.1.0.0/16
Firewall rule	Security group: All ICMP IPv4 from 10.2.0.0/16	NSG VM-1-nsg: Allow-ICMP-From-AWS, ICMP from 10.1.0.0/16, priority 310
Tunnel inside addresses	169.254.107.16/30 (link-local, used by the tunnel itself)	

Why the design looks like this

- **Non-overlapping CIDRs.** A router can only forward a packet if each destination range exists in one place. If both clouds used the same range, traffic for 10.x.x.x would never leave the local network.
- **Private subnets and no public IPs.** The goal is private connectivity. Keeping the test VMs private proves that traffic really uses the VPN, not the internet.
- **Static routing.** With one prefix on each side, static routes are simple and predictable. BGP would be preferred for many prefixes or automatic failover.
- **Route-based VPN and IKEv2.** Both AWS and Azure support route-based IKEv2 tunnels, which makes interoperability straightforward.

4. Solution Architecture

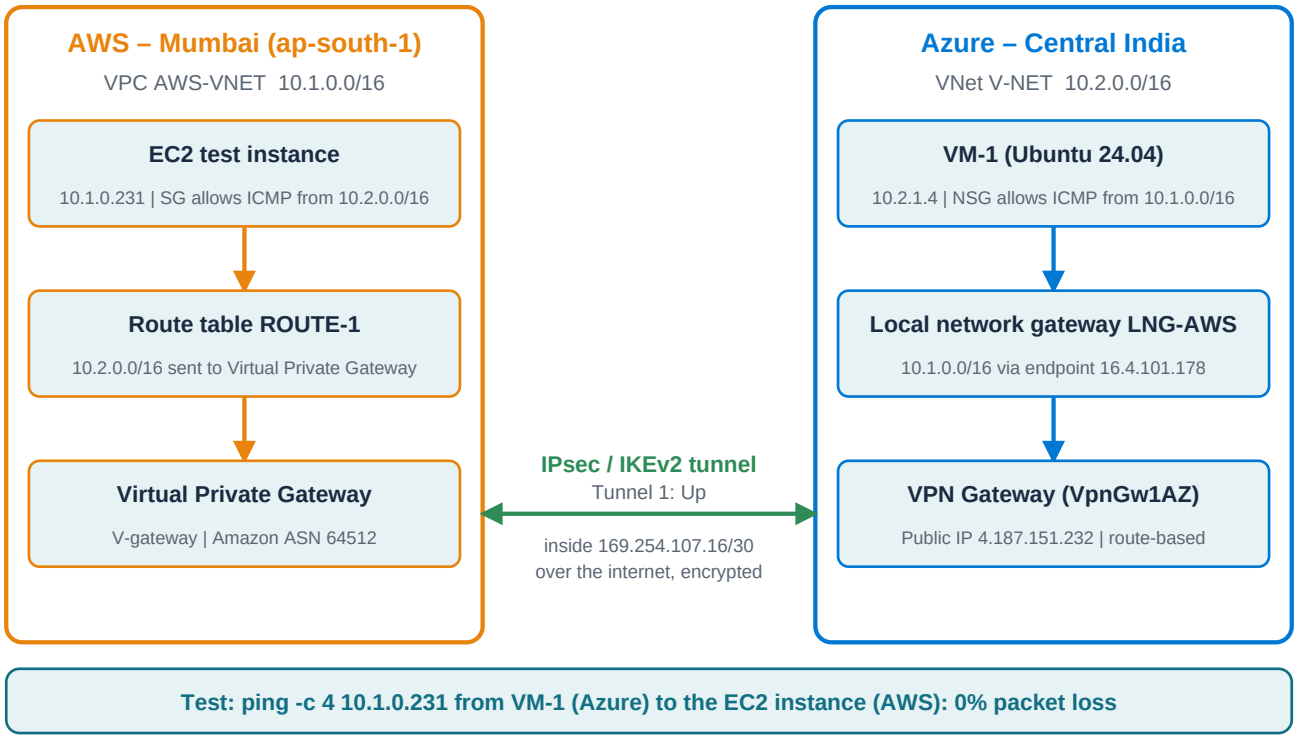


Figure 1 – Hybrid architecture: AWS VPC and Azure VNet joined by an IPsec/IKEv2 tunnel.

Packet path of the successful test (Azure to AWS)

1. VM-1 (10.2.1.4) sends an ICMP echo request to 10.1.0.231. The destination is outside V-NET, so Azure uses the route learned from the Local Network Gateway: 10.1.0.0/16 via the VPN Gateway.
2. The Azure VPN Gateway (4.187.151.232) encrypts the packet with IPsec and sends it over the internet to the AWS tunnel endpoint 16.4.101.178.
3. The AWS Virtual Private Gateway decrypts the packet and delivers it into the VPC. The EC2 security group allows ICMP from 10.2.0.0/16.
4. EC2 replies to 10.2.1.4. ROUTE-1 sends 10.2.0.0/16 to the Virtual Private Gateway, which encrypts the reply and returns it through the same tunnel.
5. The reply reaches VM-1. Both security groups and NSGs are stateful, so replies to a connection that was allowed in one direction need no extra rule.

5. Why Each Service Is Used

5.1 AWS services

Service / resource	Role in this project	Why it is needed
Amazon VPC (AWS-VNET)	Private network 10.1.0.0/16	Provides an isolated network for the EC2 instance and the address range that Azure must route to.
Subnet (Private-Subnet)	Hosts the EC2 instance	Places the instance in one Availability Zone (ap-south-1a) without public exposure.
Route table (ROUTE-1)	Route 10.2.0.0/16 to Virtual Private Gateway	Static routing is used (route propagation is off), so AWS must be told that Azure's range is reached through the VPN. Without it, replies would have no path back.
Virtual Private Gateway (V-gateway)	AWS end of the VPN, attached to the VPC	The managed VPN concentrator that terminates the IPsec tunnels and connects them to the VPC.
Customer Gateway (Azure - gateway)	Describes the Azure VPN endpoint (4.187.151.232)	Tells AWS which public IP to build the tunnel with. A BGP ASN (65000) is a required field even though BGP is not used.
Site-to-Site VPN connection (aws-azure-s2s-vpn)	Links Virtual Private Gateway and Customer Gateway	Creates the two IPsec tunnels, holds the pre-shared key and the static route 10.2.0.0/16.
EC2 instance (t3.micro)	Connectivity test target	A small, low-cost server with only a private IP; it proves that traffic arrives privately.
Security group	Inbound All ICMP IPv4 from 10.2.0.0/16	Stateful instance firewall. Opens only ping, and only from the Azure range (least privilege).

5.2 Azure services

Service / resource	Role in this project	Why it is needed
Resource group (V-NET-P)	Container for all Azure resources	Groups resources so they can be managed, billed and deleted together at clean-up.
Virtual network (V-NET) and subnets	Private network 10.2.0.0/16	Hosts VM-1 (Private-Subnet) and the mandatory GatewaySubnet used by the VPN Gateway.
Public IP (vpn, Standard)	Internet-facing address of the VPN Gateway	AWS needs a fixed public address to build the tunnel to. VpnGw1AZ gateways use Standard-SKU public IPs.
VPN Gateway (VPN, VpnGw1AZ)	Azure end of the VPN	Terminates the IPsec/IKEv2 tunnel. Route-based type with a zone-redundant SKU; active-active and BGP are disabled to keep the setup simple.
Local Network Gateway (LNG-AWS)	Represents AWS: 16.4.101.178 and 10.1.0.0/16	Tells Azure where the remote VPN endpoint is and which address range lives behind it.
Connection (LNG-AWS)	Site-to-site IPsec link	Binds the VPN Gateway to the Local Network Gateway with IKEv2, the default IPsec/IKE policy, DPD timeout 45 s and the shared key.
Virtual machine (VM-1), NIC (vm-1718)	Connectivity test source (10.2.1.4)	Runs the ping test; the NIC gives it a private-only address.
Network security group (VM-1-nsg)	Allow-ICMP-From-AWS from 10.1.0.0/16	Azure’s default rules deny inbound traffic from outside the VNet. This rule allows AWS-originated ping while everything else stays blocked.
Serial console	Terminal access to VM-1	The VM has no public IP and the SSH rule was removed, so the portal serial console is used to run the test safely.

5.3 Protocols and settings

- **IPsec with IKEv2:** IKEv2 negotiates keys and authenticates both ends; IPsec encrypts the traffic so that private data can cross the public internet.
- **Pre-shared key (PSK):** a shared secret that proves each gateway is the expected peer. It must be long, random and never published.
- **Static routing, BGP disabled:** fewer moving parts; the trade-off is manual route maintenance and no automatic failover.

6. Implementation Walkthrough

The steps below follow the build order. Each step states what was done, why, and shows the screenshot evidence. Figures are numbered in the order they appear.

Phase 1 – AWS network foundation

Step 1 – VPC, subnet and route tables

What: the VPC **AWS-VNET** (10.1.0.0/16, DNS resolution enabled, not the default VPC) contains one subnet, **Private-Subnet** in ap-south-1a, and two route tables: the main table and **ROUTE-1**, which is associated with the subnet. **Why:** this is the private network that Azure will reach; ROUTE-1 is where the Azure route is added later.

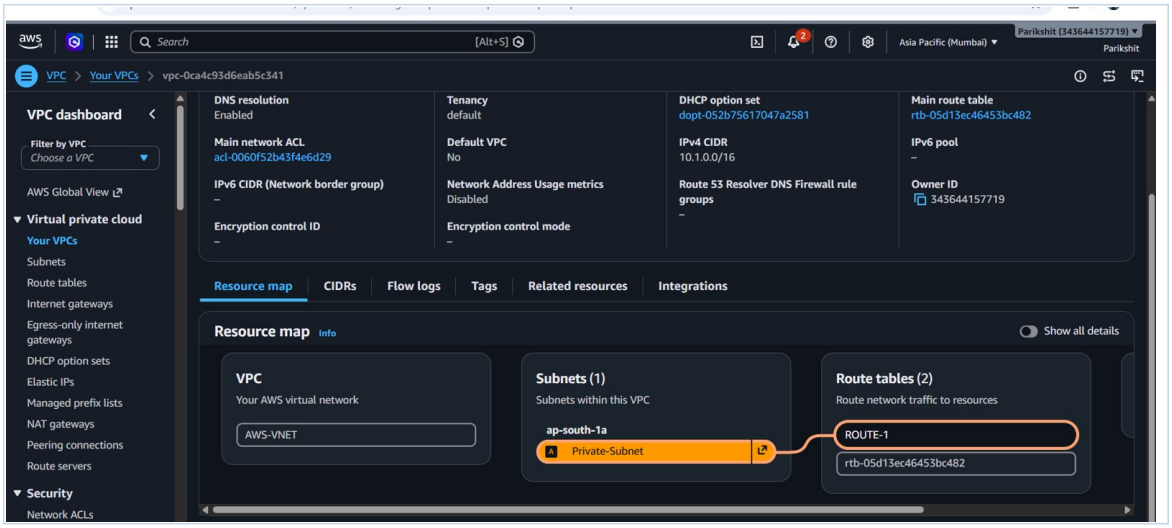


Figure 2 – VPC resource map: AWS-VNET, Private-Subnet (ap-south-1a) and route table ROUTE-1.

Step 2 – EC2 test instance

What: an EC2 instance (named EC-2) of type t3.micro in Private-Subnet, state Running, private IP **10.1.0.231**, no public IPv4 address or Elastic IP, IMDSv2 required. **Why:** it is the AWS target of the ping test; having no public IP proves the traffic is private. At the time of this screenshot no IAM role was attached (see Section 9).

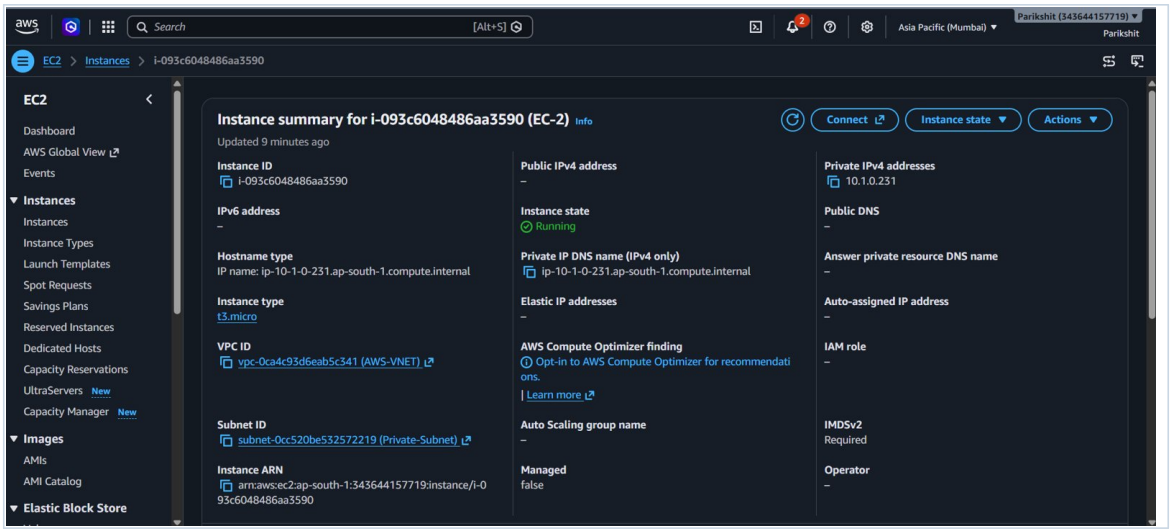


Figure 3 – EC2 instance i-093c6048486aa3590: private IP 10.1.0.231, no public IP.

Step 3 – Virtual Private Gateway

What: the Virtual Private Gateway **vgw-095bf965a9b7c5f64** (name tag V-gateway), type ipsec.1, Amazon ASN 64512, state Available and **attached** to AWS-VNET. **Why:** it is the AWS end of the VPN and the target of the route to Azure. It must be attached to the VPC before it can carry VPC traffic.

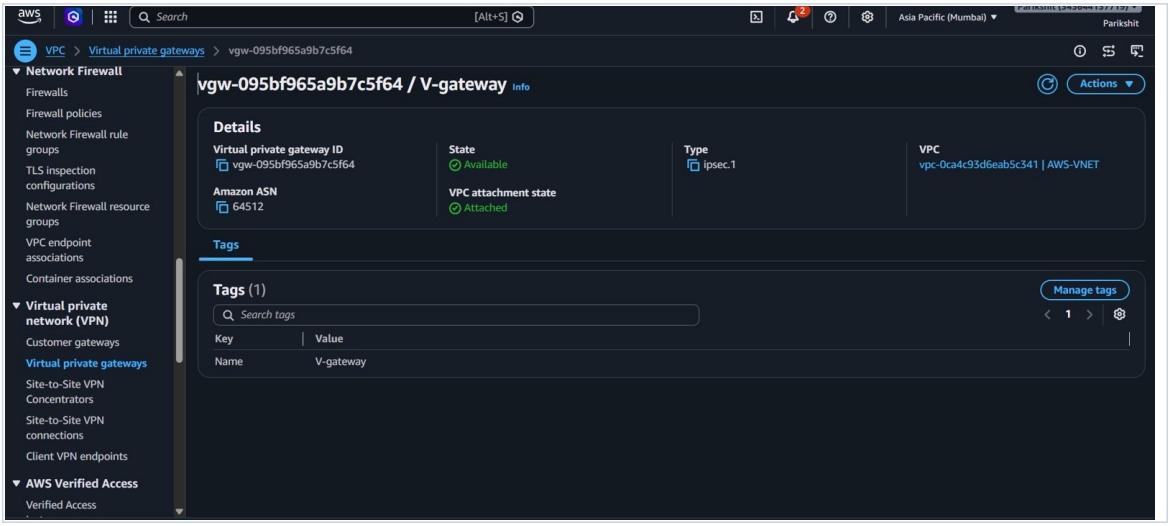


Figure 4 – Virtual Private Gateway: Available and attached to AWS-VNET.

Phase 2 – Azure network foundation

Step 4 – Azure resources in resource group V-NET-P

What: all Azure resources are in the resource group **V-NET-P**, region Central India: the VNet, VM-1 with its NIC (vm-1718), disk and NSG, the public IP *vpn*, the Virtual network gateway *VPN*, the Local network gateway *LNG-AWS* and the Connection *LNG-AWS*. **Why:** one resource group makes it easy to review the whole deployment and to delete it in one operation when the project ends.

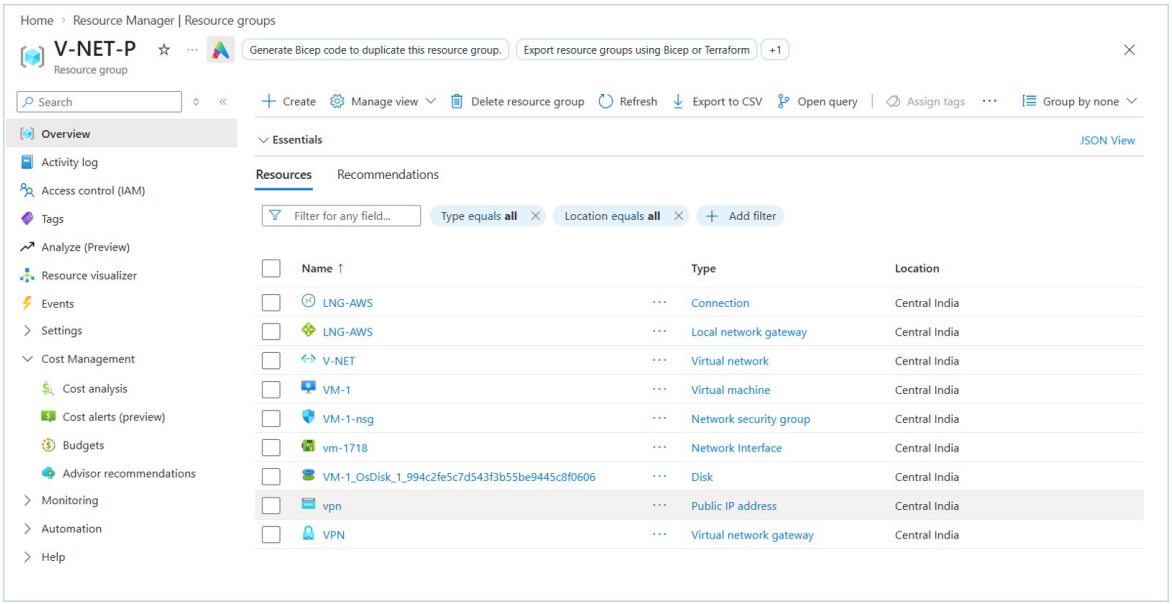


Figure 5 – Resource group V-NET-P: nine resources in Central India.

Step 5 – Test VM and its network interface

What: VM-1 is an Ubuntu 24.04 VM, size Standard D2s v3 (2 vCPU, 8 GiB), in Central India zone 1, attached to V-NET/Private-Subnet and created on 26 Oct 2026 (20:15 UTC). Its NIC **vm-1718** has the private IP **10.2.1.4** and **no public IP**, and is associated with the NSG VM-1-nsg. **Why:** VM-1 is the source of the test and mirrors the AWS instance in being private only.

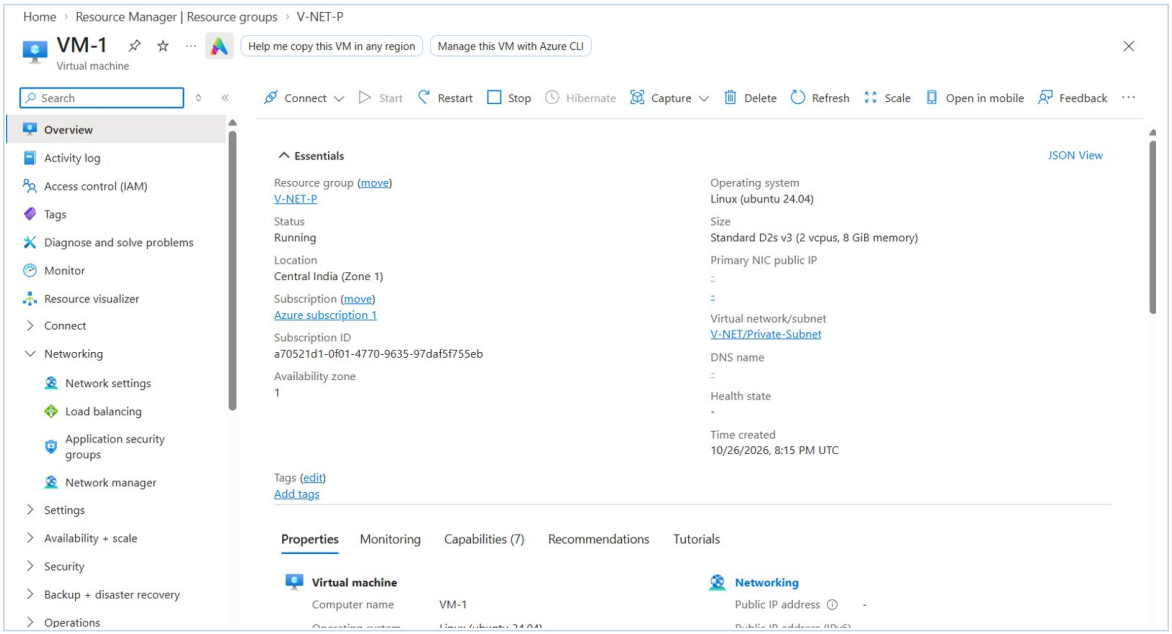


Figure 6 – VM-1 overview: Running, no public IP, V-NET/Private-Subnet.

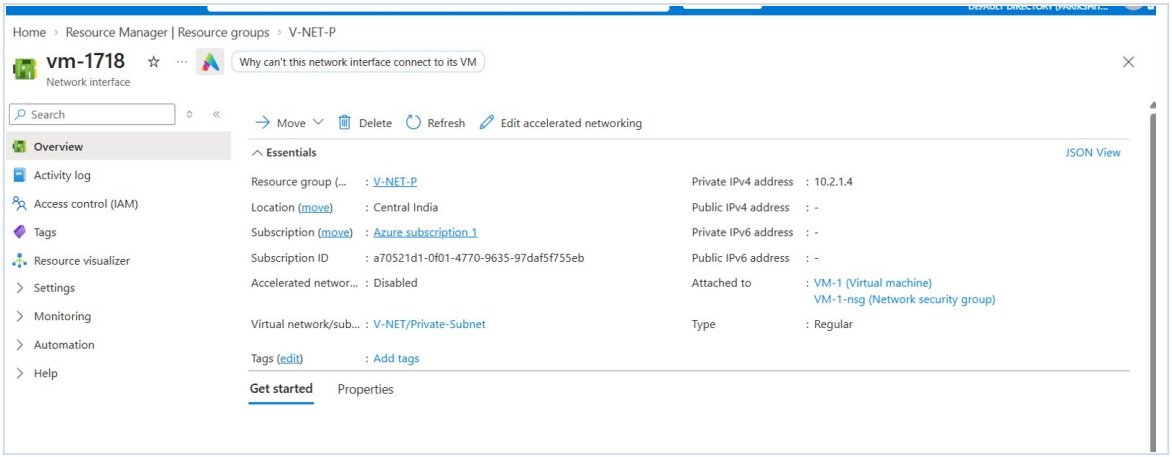


Figure 7 – NIC vm-1718: private IP 10.2.1.4, attached to VM-1 and VM-1-nsg.

Step 6 – VPN Gateway and its public IP

What: the public IP **vpn** (Standard SKU, regional, 4.187.151.232) is associated with the Virtual network gateway **VPN**. The gateway is SKU **VpnGw1AZ**, gateway type VPN, **Route-based**, in V-NET. Configuration: active-active mode *Disabled*, gateway private IPs *Disabled*, *Configure BGP* unchecked, Key Vault access disabled. **Why:** route-based IKEv2 is what AWS expects; a single instance and static routes keep the first deployment simple and cheaper. The public IP is the address AWS connects to.

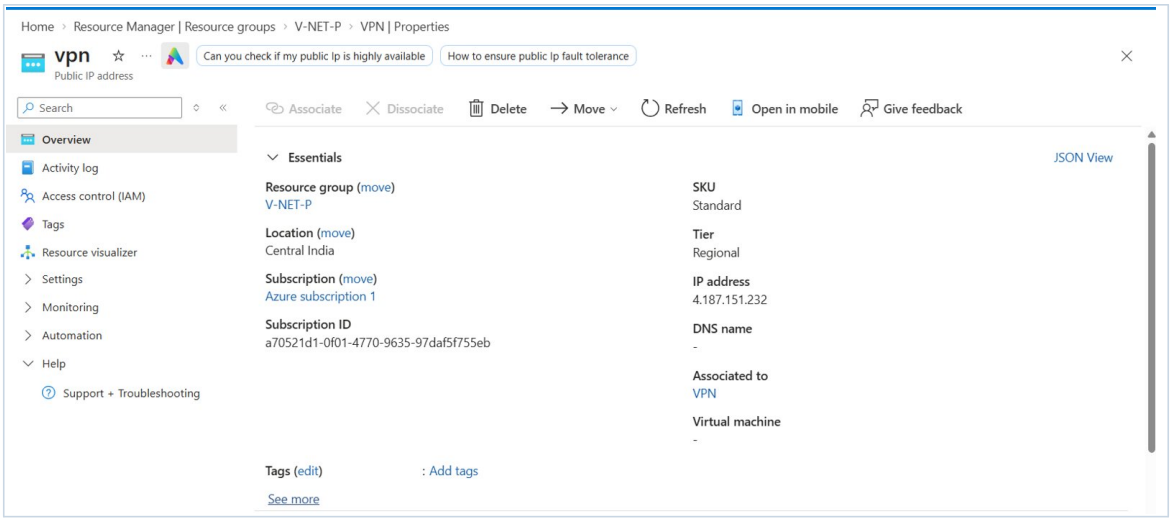


Figure 8 – Public IP "vpn": Standard SKU, 4.187.151.232, associated to the VPN gateway.

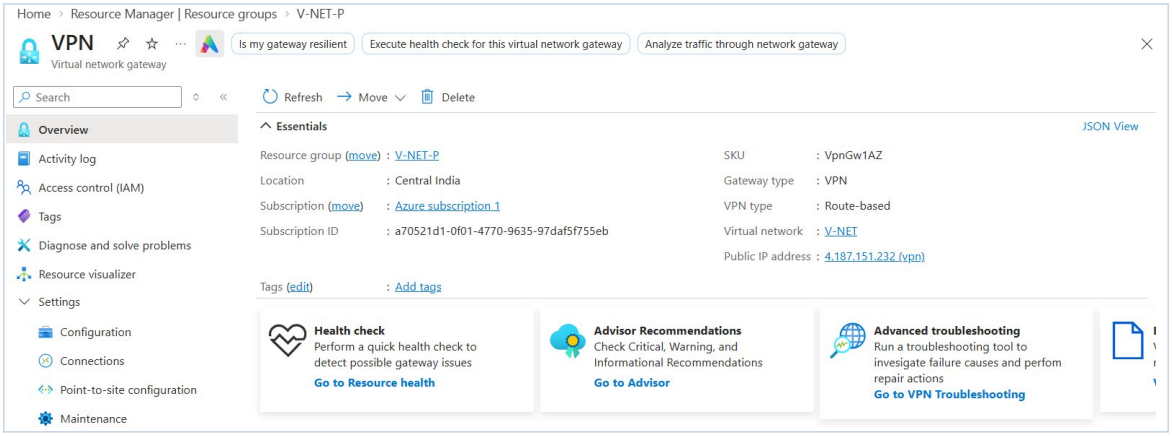


Figure 9 – VPN gateway overview: SKU VpnGw1AZ, route-based, VNet V-NET.

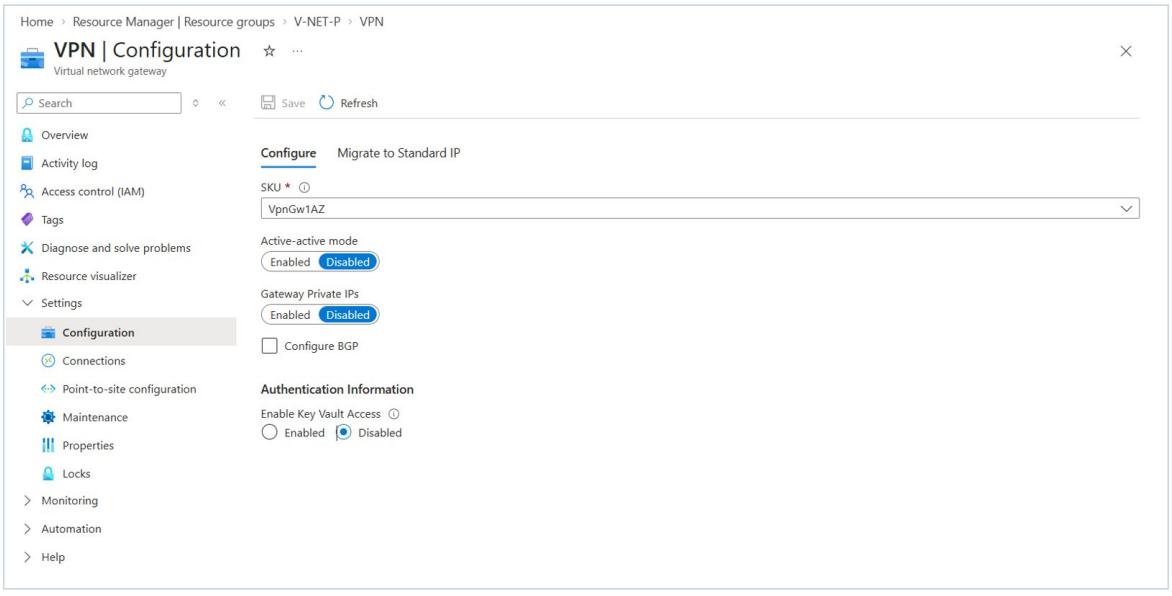


Figure 10 – VPN gateway configuration: active-active disabled, BGP off.

Phase 3 – Building the VPN between the clouds

The two clouds are configured crosswise: each side needs the public IP of the other side. The Azure gateway public IP is created first, then AWS is configured, then the AWS tunnel IP is entered in Azure.

Step 7 – AWS Customer Gateway

What: the Customer Gateway **cgw-0fcefc590762d25eb** (name tag “Azure - gateway”) with IP address **4.187.151.232**, type ipsec.1, BGP ASN 65000, state Available. **Why:** AWS must know the address of the device on the other end of the VPN; this object represents the Azure VPN Gateway.

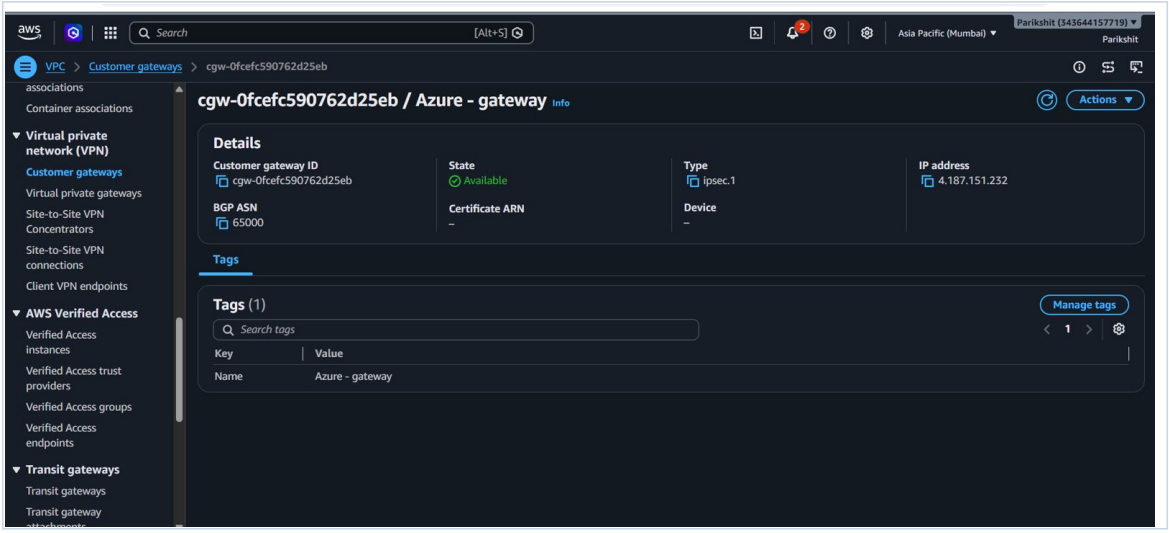


Figure 11 – Customer Gateway: Azure VPN gateway public IP 4.187.151.232.

Step 8 – AWS Site-to-Site VPN connection

What: the connection **aws-azure-s2s-vpn** (vpn-0d84293eb0704b48d) joins the Virtual Private Gateway and the Customer Gateway. It uses **static routing**, pre-shared-key authentication, standard tunnel bandwidth, acceleration disabled and a **static route 10.2.0.0/16** (the Azure VNet), state Available. **Why:** this object creates the IPsec tunnels. The static route tells the VPN which prefix lives on the Azure side.

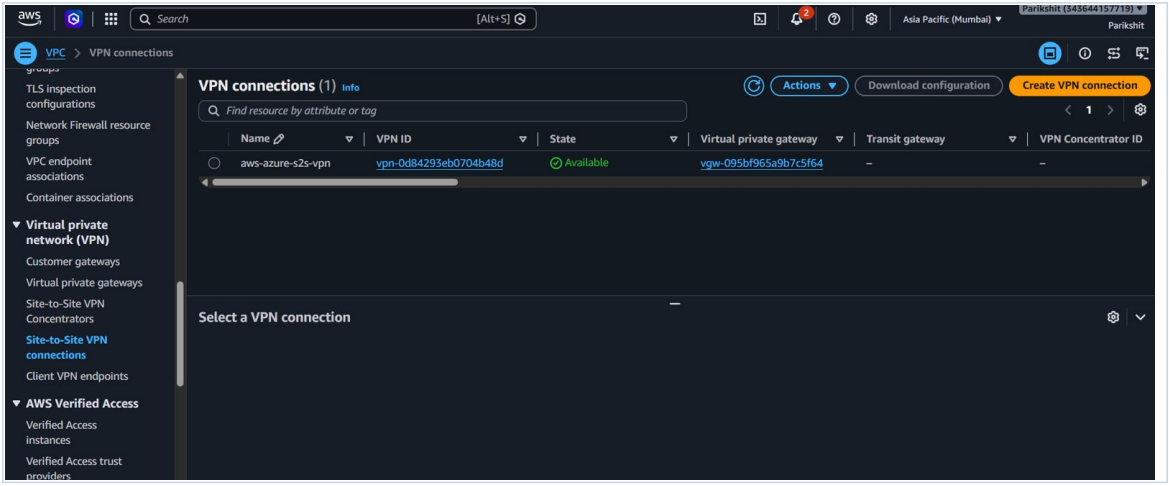


Figure 12 – VPN connections: aws-azure-s2s-vpn, state Available.

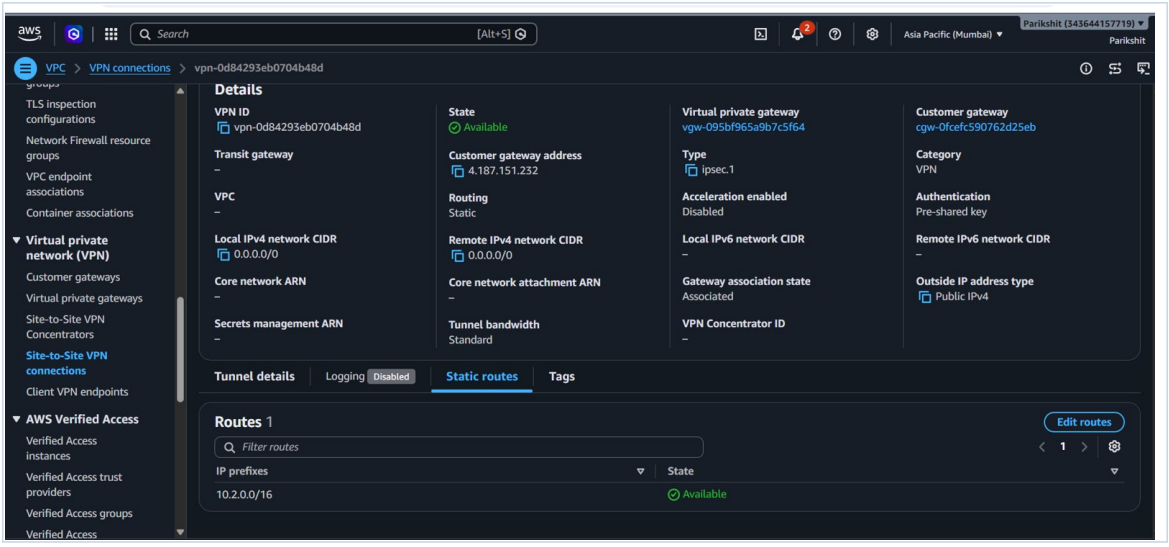


Figure 13 – VPN connection details and static route 10.2.0.0/16 (Available).

Step 9 – Azure Local Network Gateway

What: LNG-AWS was created in V-NET-P (Central India) with the endpoint IP **16.4.101.178** (the AWS Tunnel 1 outside IP) and the address space **10.1.0.0/16** (the AWS VPC); validation passed. **Why:** Azure needs a description of the remote site: where to send the tunnel, and which destination range to push into it.

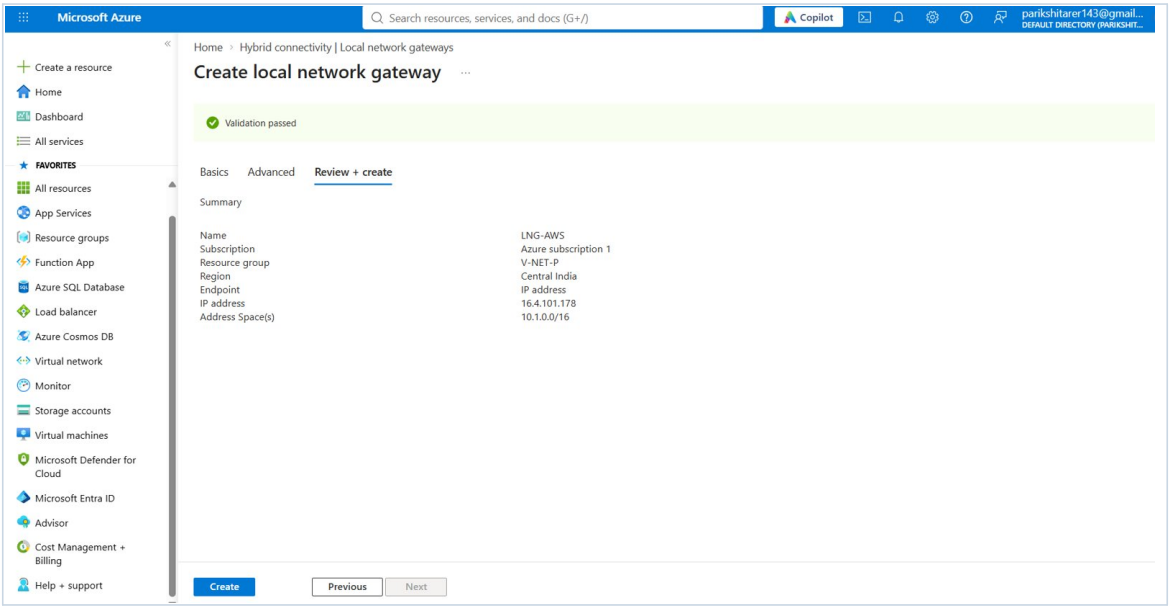


Figure 14 – Create local network gateway: endpoint 16.4.101.178, address space 10.1.0.0/16.

Step 10 – Azure VPN connection

What: a Site-to-site (IPsec) connection named **LNG-AWS** binds the VPN gateway *VPN* to the Local Network Gateway, using **IKEv2**, the default IPsec/IKE policy, policy-based traffic selectors disabled, DPD timeout 45 s and a shared key. The shared key was redacted from the screenshot. **Why:** this object starts the negotiation. The shared key must be identical to the key configured on the AWS VPN connection, otherwise the tunnel does not establish. The connection appears in the gateway's Connections list as *Connected*.

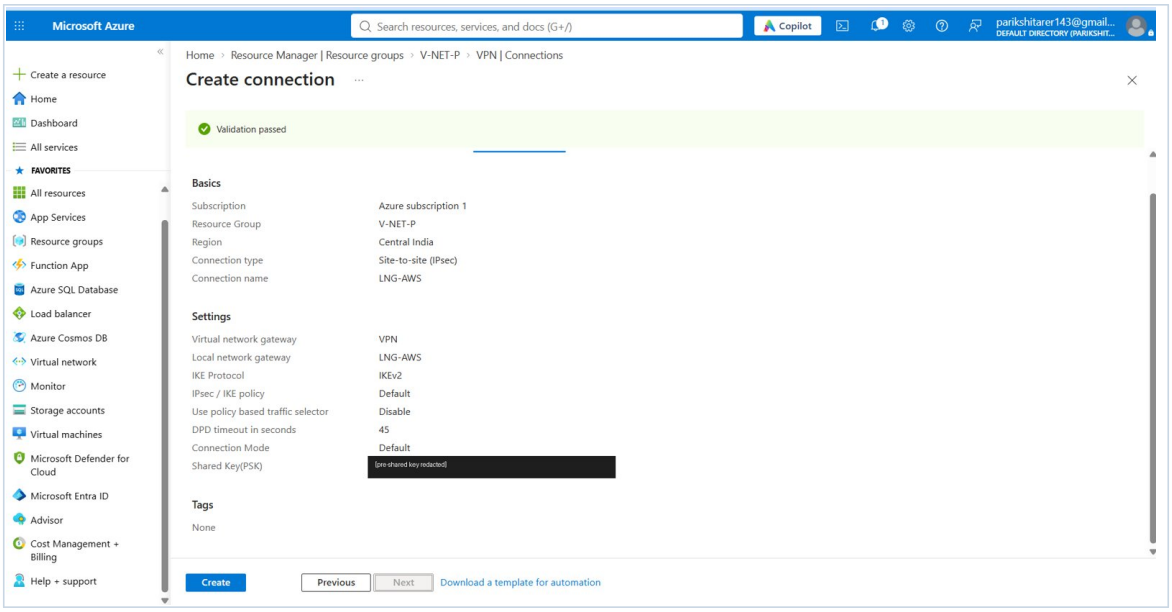


Figure 15 – Create connection review: IKEv2, default IPsec/IKE policy, DPD 45 s (pre-shared key redacted).

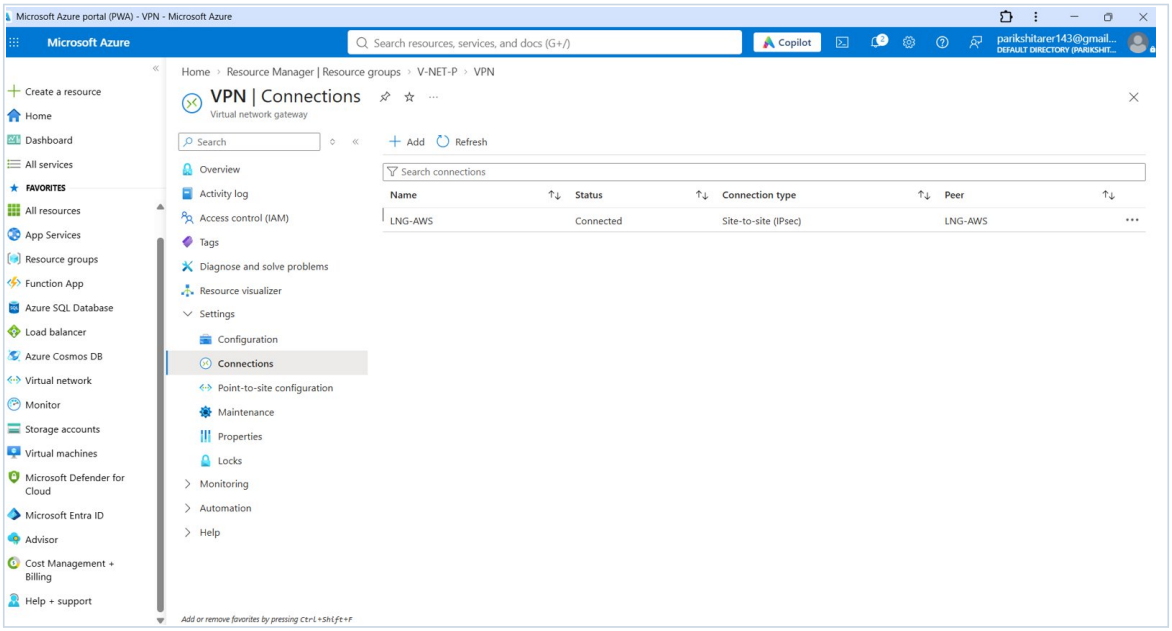


Figure 16 – VPN | Connections: LNG-AWS, status Connected, Site-to-site (IPsec).

Phase 4 – Routing and security rules

Step 11 – AWS route to Azure

What: in route table **ROUTE-1** (rtb-09a896243360c52d7) the existing *local* route 10.1.0.0/16 is kept and a new route was added: destination **10.2.0.0/16**, target **Virtual Private Gateway vgw-095bf965a9b7c5f64**. The screenshot is the Edit routes page before saving; the successful ping confirms the route was applied. **Why:** because routes are static, AWS does not learn the Azure range automatically. Without this route, the EC2 reply to 10.2.1.4 would have no path back.

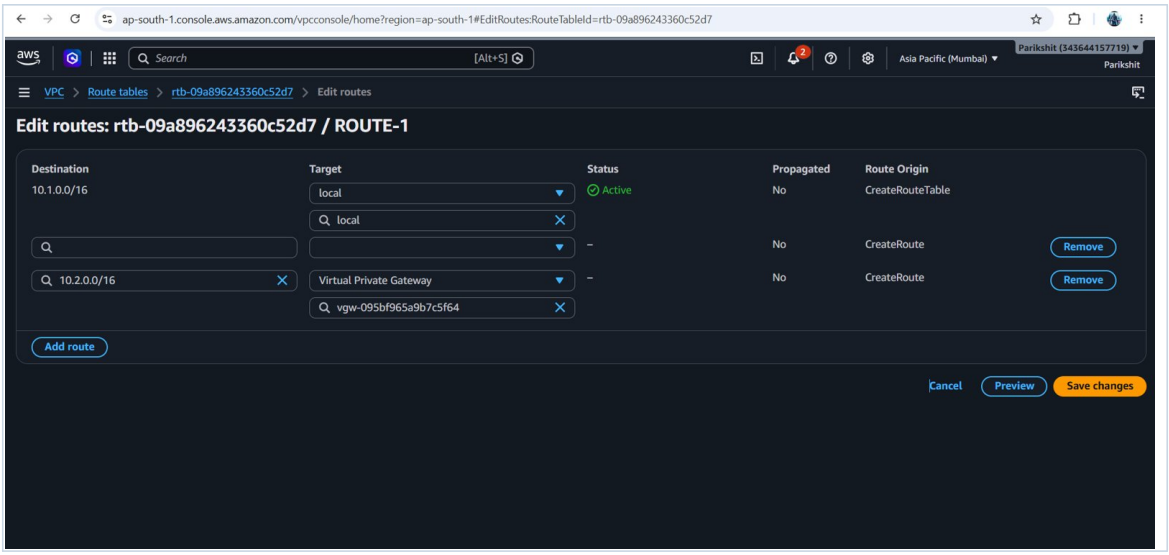


Figure 17 – ROUTE-1: new route 10.2.0.0/16 to the Virtual Private Gateway.

Step 12 – AWS security group rule

What: an inbound rule on the instance's security group (sg-0aa4b1252480a775b): type **All ICMP – IPv4**, source **10.2.0.0/16**, with a short description. **Why:** security groups deny inbound traffic by default; this rule opens only ping, only from the Azure range.

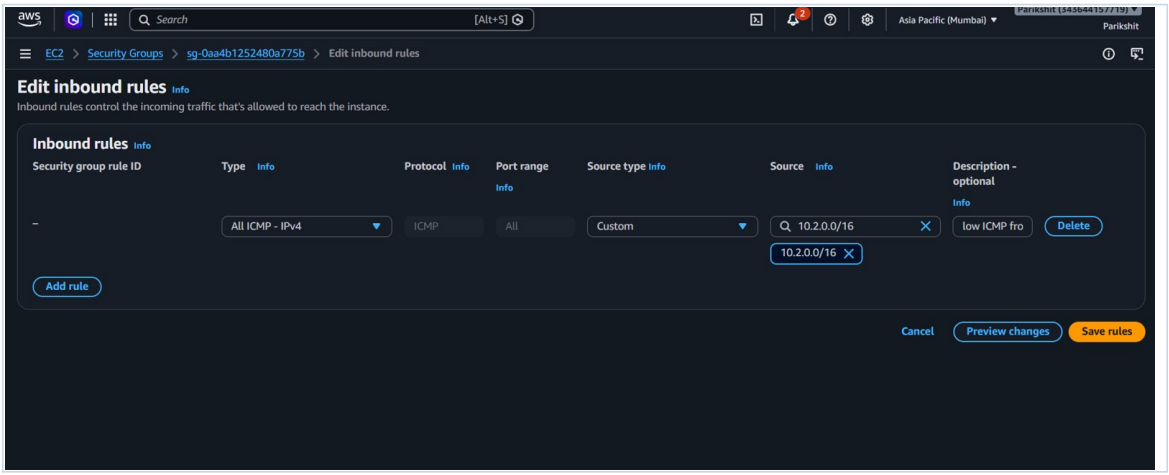


Figure 18 – Security group inbound rule: All ICMP IPv4 from 10.2.0.0/16.

Step 13 – Azure NSG rule

What: the NSG **VM-1-nsg** (attached to the NIC, not the subnet) has one custom inbound rule, **Allow-ICMP-From-AWS**: priority **310**, protocol ICMP, any port, source **10.1.0.0/16**, destination any, action Allow; the default rules (AllowVnetInBound, AllowAzureLoadBalancerInBound, DenyAllInBound) remain. The portal notification shows that the earlier **SSH** rule was deleted. **Why:** the rule lets AWS-initiated ping reach VM-1 in the future reverse test; removing SSH reduces the attack surface because access uses the serial console.

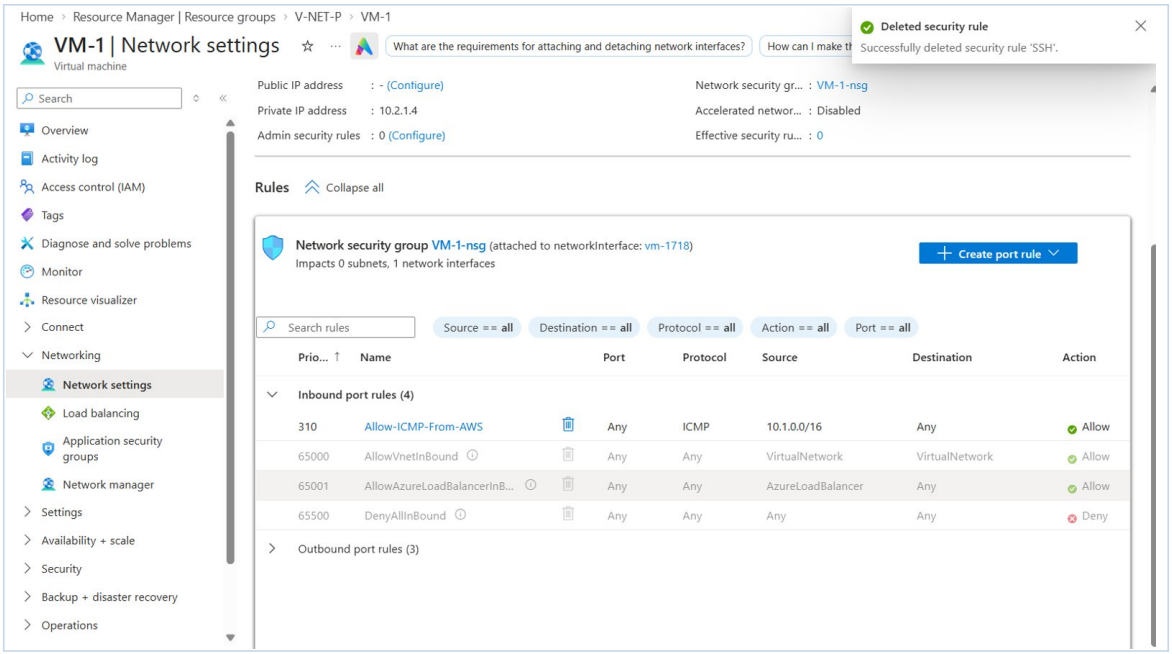


Figure 19 – VM-1 network settings: Allow-ICMP-From-AWS (priority 310) and notice that the SSH rule was deleted.

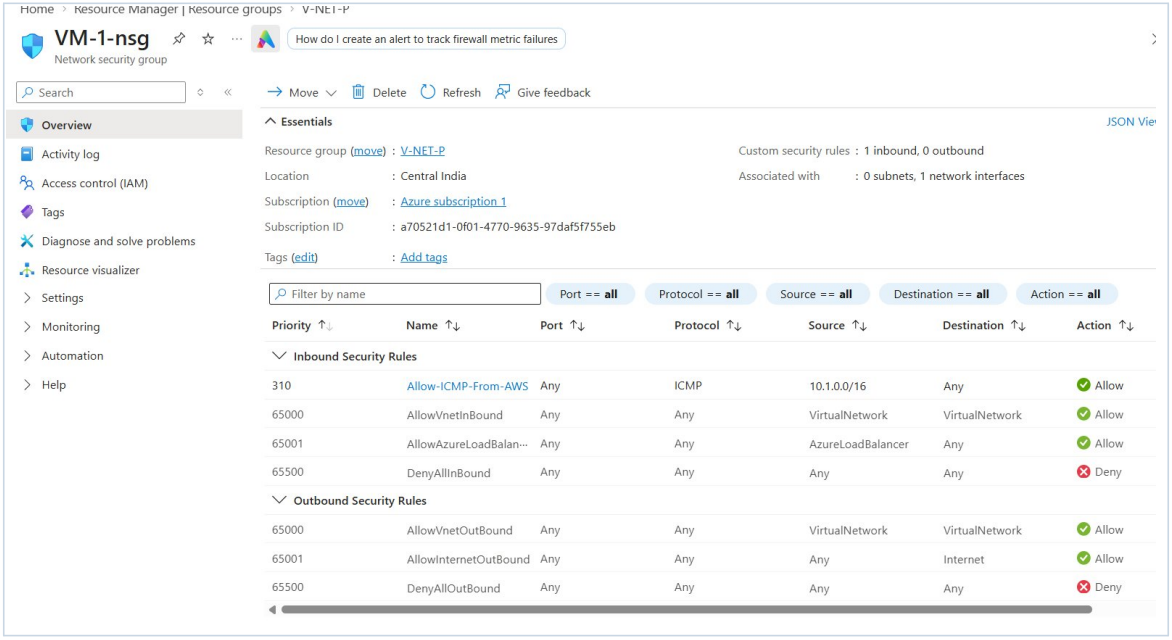


Figure 20 – NSG VM-1-nsg overview: 1 custom inbound rule, associated with 1 network interface.

Phase 5 – Verification

Step 14 – Tunnel and connection status

On AWS, the VPN connection shows **Tunnel 1: Up** (outside IP 16.4.101.178, inside CIDR 169.254.107.16/30, provisioning status Available, last status change 27 Oct 2026, 01:33:06 IST). On Azure, the connection shows **Connected**, with about 1008 B of data in and 1008 B out. AWS provides two tunnels per VPN connection; only Tunnel 1 is visible in the screenshot and it is the only one configured on the Azure side. **Why this matters:** a connected state proves that the IPsec tunnel is established. It does not by itself prove that applications or every network path work, which is why the ping test follows.

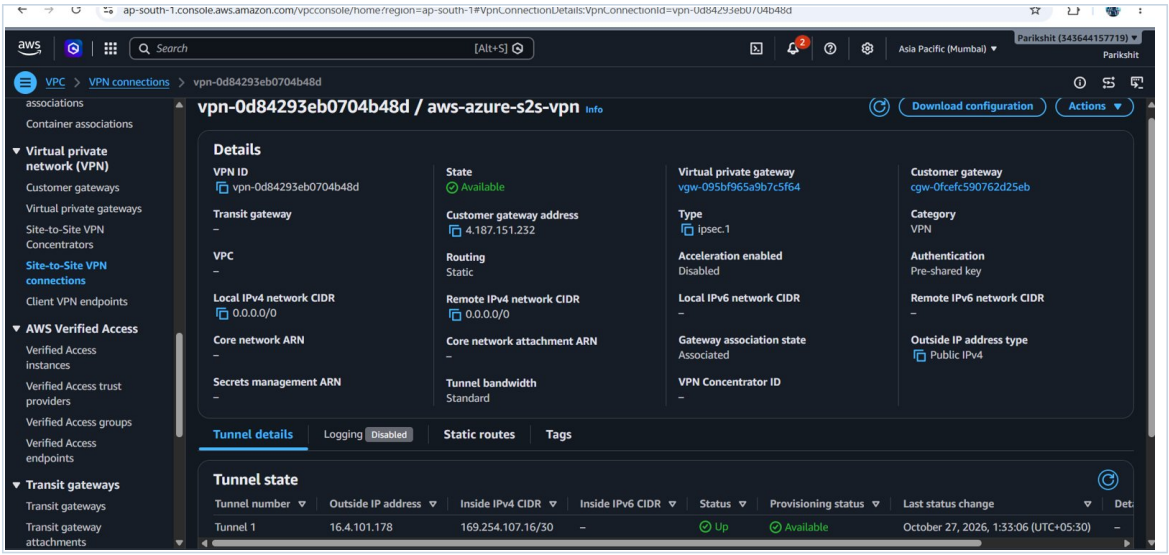


Figure 21 – AWS VPN connection: Tunnel 1 Up, routing static, authentication pre-shared key.

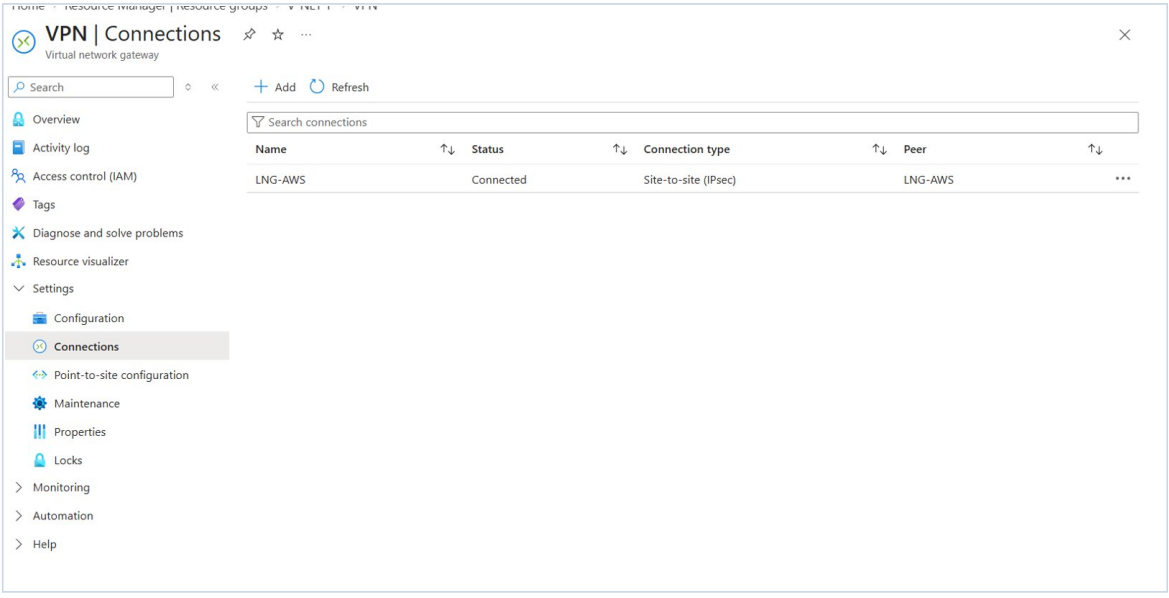


Figure 22 – Azure VPN gateway → Connections: LNG-AWS Connected.

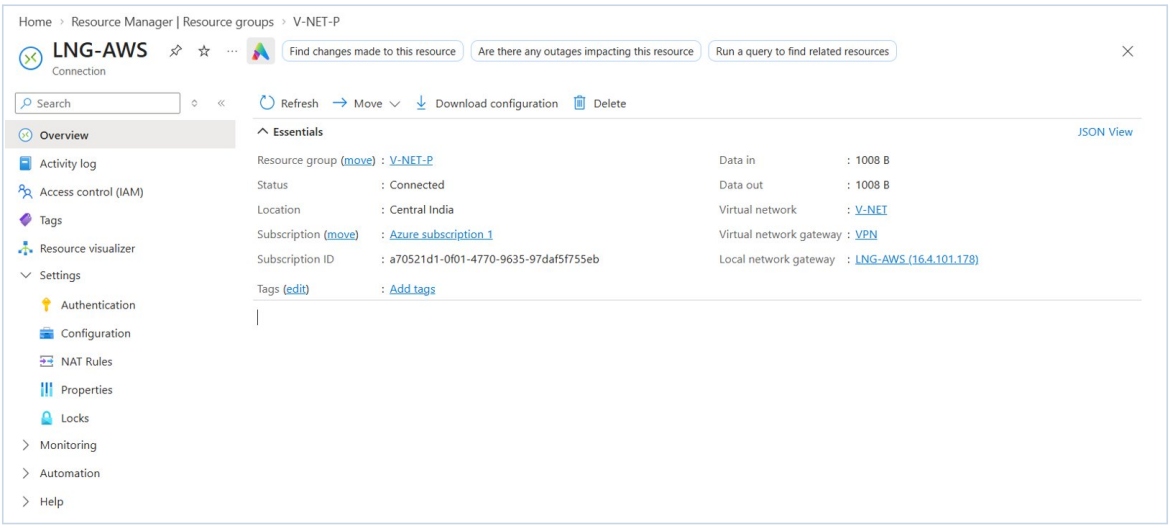


Figure 23 – Azure connection overview: Connected, data in/out 1008 B, peer LNG-AWS (16.4.101.178).

The gateway metrics for the last hour show two short bursts of a few hundred bytes in both tunnel ingress and egress, consistent with the small ICMP test traffic and the times of the two ping runs.

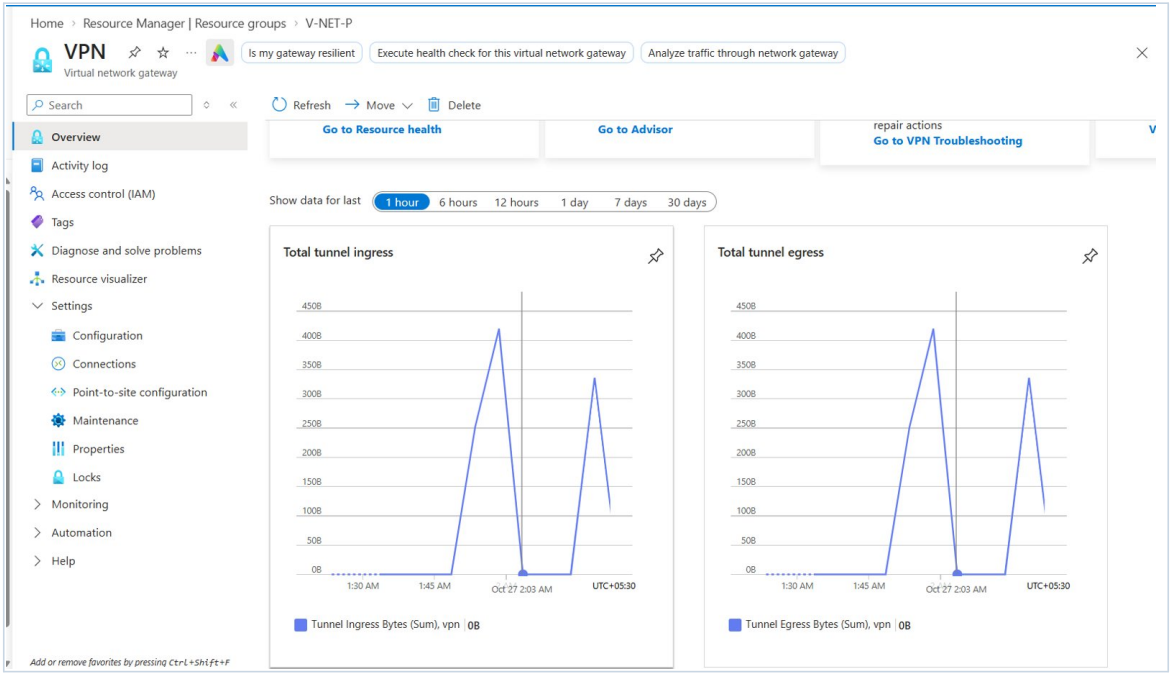


Figure 24 – VPN gateway metrics: total tunnel ingress and egress (last hour).

Step 15 – Connectivity test: Azure VM to AWS EC2

From the Azure serial console of VM-1 the command `ping -c 4 10.1.0.231` was run. The replies came from the private address of the EC2 instance with TTL 64 and a round-trip time of roughly 6–7 ms, which is plausible for two regions in western India.

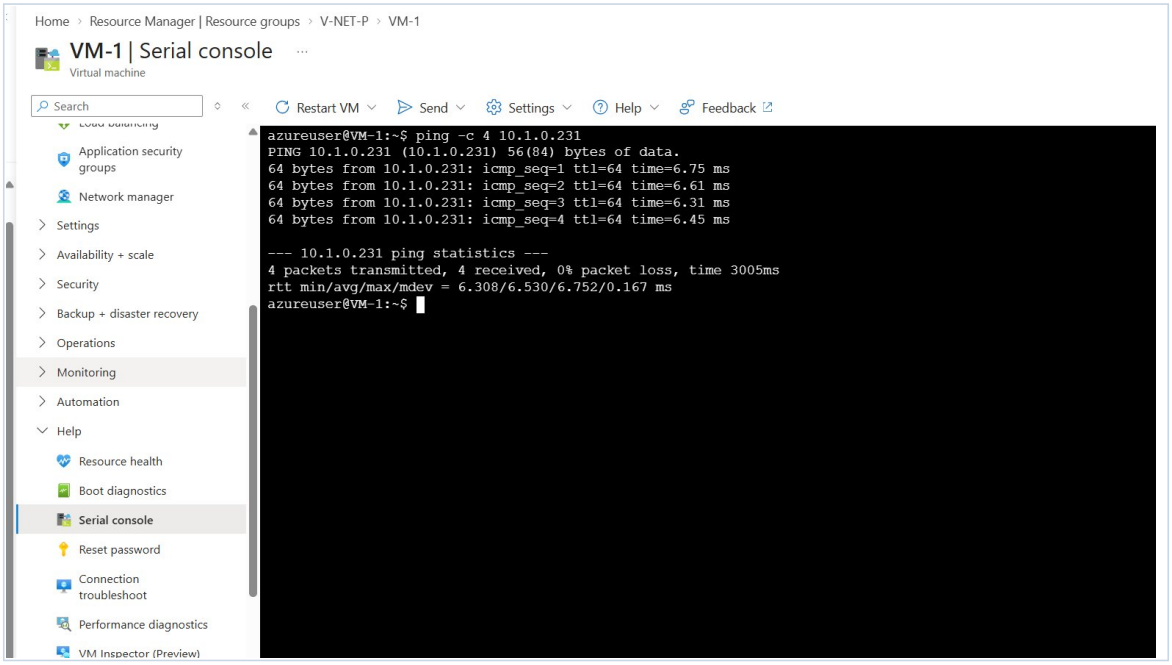


Figure 25 – First ping run from VM-1: 4 sent, 4 received, 0% packet loss.

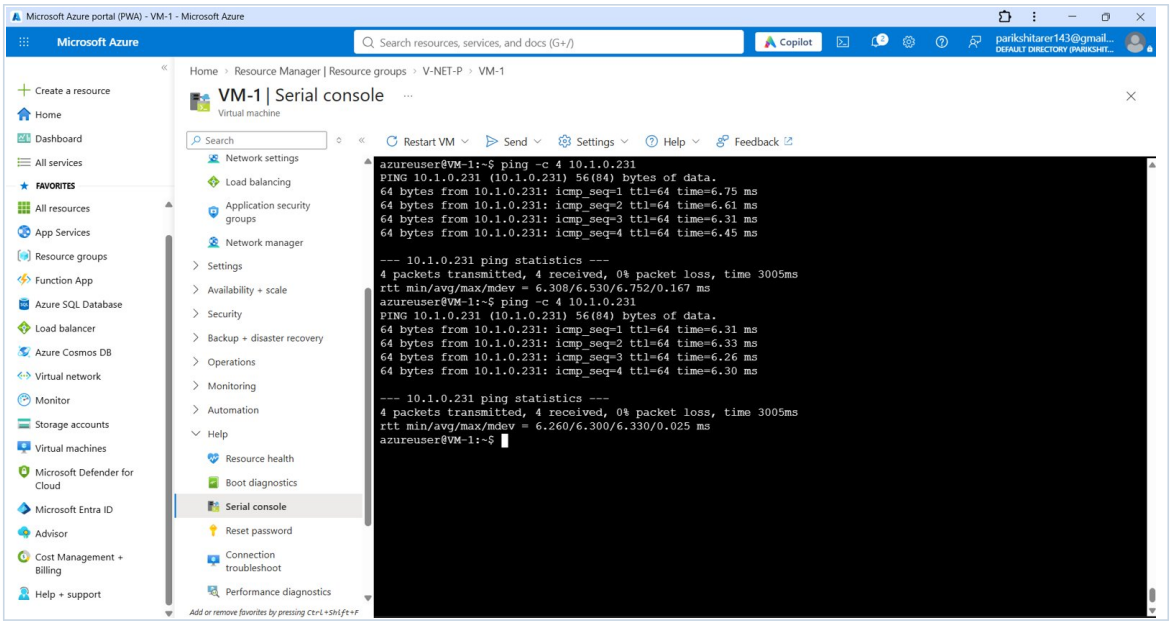


Figure 26 – Console showing two consecutive ping runs, both with 0% packet loss.

Metric	Run 1	Run 2
Source / destination	VM-1 (10.2.1.4) to EC2 (10.1.0.231)	Same
Packets sent / received	4 / 4	4 / 4
Packet loss	0%	0%
RTT min / avg / max / mdev	6.308 / 6.530 / 6.752 / 0.167 ms	6.260 / 6.300 / 6.330 / 0.025 ms
Total time	3005 ms	3005 ms

What the test proves

- VM-1 can reach the EC2 instance by its private IP across the VPN.
- The route, tunnel and the AWS security-group rule work for this path (the reply returns through the same tunnel).
- Neither test machine needed a public IP address.
- **Not proven:** AWS-to-Azure connectivity, other protocols and ports, or failover, because these were not tested.

7. Results Summary

Check	Result	Evidence
AWS Virtual Private Gateway attached	Pass	Figure 4
AWS Customer Gateway available	Pass	Figure 11
AWS VPN connection available, static route 10.2.0.0/16	Pass	Figures 12, 13
Azure VPN Gateway (VpnGw1AZ) created with public IP	Pass	Figures 8, 9, 10
Local Network Gateway and connection created	Pass	Figures 14, 15, 16
AWS route to Azure range	Pass	Figure 17
AWS Tunnel 1 Up	Pass	Figure 21
Azure connection Connected	Pass	Figures 22, 23
Azure to AWS private ping, 0% loss	Pass	Figures 25, 26
AWS to Azure private ping	Not tested	–
Application (HTTP) test across clouds	Not implemented	–
Second tunnel / redundancy	Not implemented	Only Tunnel 1 configured in Azure

8. Project Notes Versus Screenshots

The project notes and the console screenshots differ in a few details. This report follows the screenshots, and the differences are listed here so that the documentation stays accurate.

Item	Project notes	Screenshots
Azure connection name	Connection-AWS	Connection is named LNG-AWS (same as the Local Network Gateway)
AWS Customer Gateway name	azure-customer-gateway	Name tag “Azure - gateway” (cgw-0fcefc590762d25eb)
Virtual Private Gateway name	not named	Name tag V-gateway
NSG rule priority	300	310
Ping result	Average 6.479 ms (6.306 / 6.479 / 6.857 / 0.225)	Two captured runs: 6.530 ms and 6.300 ms average; the 6.479 ms run is not in the screenshots
EC2 IAM role	EC2-SSM-Role attached	No IAM role shown on the instance page (screenshot may predate attachment)
EC2 instance name	not named	Name tag EC-2

9. Troubleshooting and Pending Work

9.1 Session Manager access to the EC2 instance

Session Manager stayed disconnected, so the reverse ping was skipped. Likely causes, based on the screenshots: the instance sits in a private subnet with no public IP and ROUTE-1 has no internet or NAT route, and an IAM instance profile with SSM permissions was not shown on the instance. The SSM agent therefore cannot reach the Systems Manager service. To fix it, attach a role with the *AmazonSSMManagedInstanceCore* policy and either add VPC interface endpoints for `ssm`, `ssmmessages` and `ec2messages`, or add a NAT gateway. After a few minutes the instance should register as a managed node.

9.2 General VPN troubleshooting checklist

- Tunnel Down: compare the pre-shared key, IKE version (IKEv2) and the peer public IPs on both sides.
- Tunnel Up but no ping: check the route in ROUTE-1, the Local Network Gateway address space, the security group and the NSG, and the VM’s own firewall.
- Ping works one way only: check the return route and the rule on the destination side.
- Use the AWS tunnel status, Azure connection status and gateway metrics (Figures 21–24) to see if traffic flows.

9.3 Pending items

Enhancement	Status	Why it matters
Azure-to-AWS private ping	Completed	Core requirement
AWS-to-Azure private ping	Not tested	Proves the return direction and the Azure NSG rule
HTTP application test	Not implemented	Shows that real applications work, not just ICMP
Second tunnel / redundancy	Not implemented	Removes the single point of failure; configure Tunnel 2 or use active-active with BGP
Infrastructure as Code	Not implemented	Repeatable deployment with Terraform or Bicep
Cost clean-up	Pending decision	Gateways are billed while they exist

10. Findings and Recommendations

#	Observation	Recommendation
1	Only AWS Tunnel 1 is configured on the Azure side.	Add the second AWS tunnel (second Local Network Gateway and connection) or move to an active-active gateway with BGP for resilience.
2	The AWS VPN connection shows Local/Remote IPv4 CIDR 0.0.0.0/0.	Optionally narrow to 10.2.0.0/16 and 10.1.0.0/16 so that only the intended ranges are tunnelled.
3	ICMP rules allow all ICMP types from the whole peer range, with any port in the NSG.	Keep for testing only; afterwards remove them or restrict to the ports and protocols the application needs.
4	VPN logging is disabled on the AWS connection.	Enable tunnel logging (CloudWatch) and set up Azure Monitor alerts on the connection status.
5	The create-connection screen displayed the pre-shared key and a screenshot of it was shared outside the portal.	Rotate the PSK on both sides, use a long random key, store it in a secret manager and never include it in documentation or a public repository.
6	Reverse (AWS to Azure) testing is missing; Session Manager does not connect.	Apply the fix in Section 9.1 and run `ping -c 4 10.2.1.4` from the EC2 instance.
7	Resource names differ between the notes and the console.	Use one naming convention (for example vpn-aws-azure, cgw-azure) and update the README.
8	Screenshots show account, subscription and tunnel IP details.	Blur identifiers before publishing; use the repository README for architecture, not raw console captures.

11. Cost Management and Clean-up

VPN gateways are billable resources even when little traffic flows. Potential charges: the Azure VPN Gateway (VpnGw1AZ) and its Standard public IP, the AWS Site-to-Site VPN connection, the Azure VM and managed disk, the EC2 instance and its storage, and data transfer between the clouds. Deallocating the Azure VM does not stop VPN gateway charges. Check current pricing for Central India and Mumbai.

Suggested clean-up order when the project is finished:

1. Delete the Azure connection (LNG-AWS), then the Local Network Gateway.
2. Delete the Azure VPN Gateway (this takes several minutes) and its public IP.
3. Delete the AWS Site-to-Site VPN connection, then the Customer Gateway, then detach and delete the Virtual Private Gateway.
4. Remove the 10.2.0.0/16 route from ROUTE-1 and the ICMP rules.
5. Terminate the EC2 instance and delete the Azure VM, or delete the whole resource group V-NET-P.
6. Keep the VPC and VNet only if they will be reused.

12. What Was Learned

- **Hybrid networking:** connecting two separate clouds into one private address space.
- **Site-to-Site VPN:** gateway, remote-site object, connection and shared key on each side.
- **Routing:** why static routes are needed on both sides and how non-overlapping CIDRs make routing possible.
- **Network security:** security groups and NSGs as stateful firewalls with least-privilege rules.

- **Troubleshooting:** reading tunnel status, connection state, metrics and ping output; understanding why private instances cannot reach Systems Manager without endpoints.

13. Resume-Ready Description

AWS–Azure Hybrid Cloud Connectivity Using Site-to-Site VPN
Technologies: Amazon VPC, EC2, AWS Site-to-Site VPN, Virtual Private Gateway, Azure Virtual Network, Azure VPN Gateway, IPsec, IKEv2, Security Groups, NSG.

- Built a hybrid network connecting an AWS VPC (10.1.0.0/16, Mumbai) and an Azure VNet (10.2.0.0/16, Central India) with a Site-to-Site IPsec/IKEv2 VPN.
- Configured the Virtual Private Gateway, Customer Gateway, route-based Azure VPN Gateway, Local Network Gateway and static routes between non-overlapping address spaces.
- Deployed private test instances in both clouds and restricted traffic with a security group and an NSG limited to ICMP between the two ranges.
- Validated Azure-to-AWS private connectivity with 0% packet loss and about 6.3–6.5 ms latency; analysed tunnel status, routing and access issues.

14. Conclusion

The project successfully builds and verifies private connectivity between AWS and Azure. The Site-to-Site IPsec tunnel is up on both sides, routes and security rules are in place, and VM-1 reaches the EC2 instance by private IP with no packet loss. Each service has a defined role: the VPC/VNet and subnets provide the networks, the gateways terminate the tunnel, the Customer and Local Network Gateways describe the remote site, routes and security rules control traffic, and the test VMs prove the result. The next steps are reverse testing, an HTTP application test, a second tunnel, Infrastructure as Code and clean-up of the billable gateways.

Appendix A – Resource Inventory

Cloud	Resource	Name / ID	Key values
AWS	VPC	AWS-VNET, vpc-0ca4c93d6eab5c341	10.1.0.0/16, DNS resolution enabled
AWS	Subnet	Private-Subnet, subnet-0cc520be532572219	ap-south-1a
AWS	Route tables	ROUTE-1 (rtb-09a896243360c52d7); main rtb-05d13ec46453bc482	10.2.0.0/16 to vgw-095bf965a9b7c5f64
AWS	Virtual Private Gateway	V-gateway, vgw-095bf965a9b7c5f64	ASN 64512, attached
AWS	Customer Gateway	Azure - gateway, cgw-0fcefc590762d25eb	4.187.151.232, ASN 65000
AWS	VPN connection	aws-azure-s2s-vpn, vpn-0d84293eb0704b48d	Static, PSK, Tunnel 1 16.4.101.178 Up
AWS	EC2 instance	EC-2, i-093c6048486aa3590	t3.micro, 10.1.0.231, no public IP
AWS	Security group	sg-0aa4b1252480a775b	All ICMP IPv4 from 10.2.0.0/16
Azure	Resource group	V-NET-P	Central India
Azure	Virtual network	V-NET	10.2.0.0/16
Azure	VPN Gateway	VPN	VpnGw1AZ, route-based, active-active off, BGP off
Azure	Public IP	vpn	4.187.151.232, Standard, regional
Azure	Local Network Gateway	LNG-AWS	16.4.101.178; 10.1.0.0/16
Azure	Connection	LNG-AWS	Site-to-site IPsec, IKEv2, Connected
Azure	Virtual machine	VM-1	Ubuntu 24.04, D2s v3, 10.2.1.4, zone 1
Azure	NIC / NSG	vm-1718 / VM-1-nsg	Allow-ICMP-From-AWS, priority 310

Appendix B – Verification Commands

Commands that can be used to repeat or extend the verification. The CLI commands are provided for reference and were not run in this project.

```
# From VM-1 (Azure serial console)
ping -c 4 10.1.0.231

# From the EC2 instance (after Session Manager works) - reverse test
ping -c 4 10.2.1.4

# Azure: connection status
az network vpn-connection show -g V-NET-P -n LNG-AWS --query connectionStatus -o tsv

# AWS: tunnel status
aws ec2 describe-vpn-connections --region ap-south-1 \
  --vpn-connection-ids vpn-0d84293eb0704b48d \
  --query "VpnConnections[0].VgwTelemetry[].[OutsideIpAddress,Status]" --output table
```

Appendix C – Figure Index

Figure	Content	Cloud	Captured (IST, 27 Oct)
Figure 2	VPC resource map	AWS	02:16
Figure 3	EC2 instance	AWS	02:17
Figure 4	Virtual Private Gateway	AWS	02:17
Figure 5	Resource group V-NET-P	Azure	02:19
Figure 6	VM-1 overview	Azure	02:21
Figure 7	NIC vm-1718	Azure	02:23
Figure 8	Public IP vpn	Azure	02:23
Figure 9	VPN gateway overview	Azure	02:22
Figure 10	VPN gateway configuration	Azure	02:22
Figure 11	Customer Gateway	AWS	02:17
Figure 12	VPN connections list	AWS	02:20
Figure 13	VPN details and static route	AWS	02:20
Figure 14	Create local network gateway	Azure	01:24
Figure 15	Create connection (PSK redacted)	Azure	01:30
Figure 16	VPN | Connections	Azure	01:35
Figure 17	Edit routes ROUTE-1	AWS	01:37
Figure 18	Security group inbound rule	AWS	01:48
Figure 19	VM-1 network settings	Azure	01:51
Figure 20	NSG overview	Azure	02:26
Figure 21	VPN tunnel details	AWS	02:16
Figure 22	VPN | Connections (Connected)	Azure	02:22
Figure 23	Connection overview	Azure	02:22
Figure 24	Gateway metrics	Azure	02:21
Figure 25	Ping run 1	Azure	01:55
Figure 26	Ping runs 1 and 2	Azure	02:18

Appendix D – Glossary

Term	Meaning
Site-to-Site VPN	Encrypted tunnel that connects two private networks across the internet.
IPsec / IKEv2	Protocol suite that encrypts traffic (IPsec) and negotiates keys and peer identity (IKEv2).
Virtual Private Gateway	AWS-managed VPN endpoint attached to a VPC.
Customer Gateway	AWS object that represents the remote VPN device (here the Azure VPN gateway).
VPN Gateway (Azure)	Azure-managed VPN endpoint deployed in a GatewaySubnet of a VNet.
Local Network Gateway	Azure object that represents the remote site: its VPN IP and address prefixes.
Static routing / BGP	Manually configured routes versus automatic route exchange between gateways.
Security group / NSG	Stateful virtual firewalls for AWS instances / Azure subnets and network interfaces.
CIDR	Notation for an IP address range, for example 10.1.0.0/16.
ICMP	Protocol used by ping.